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***Abstract*—The abstract**

***Index Terms*—IEEEtran, journal, LaTeX, paper, template.**

I. Introduction

Notation: For any $W = W^\top \geq 0$, denote by

$$\lambda^{\uparrow}(W) = \mathrm{col}\{\lambda_1^{\uparrow}(W), \ldots, \lambda_{nq}^{\uparrow}(W)\} \tag{1}$$

the eigenvalues of W arranged in nondecreasing order, i.e.,

$$0 \leq \lambda_1^{\uparrow}(W) \leq \lambda_2^{\uparrow}(W) \leq \cdots \leq \lambda_{nq}^{\uparrow}(W).$$

II. Problem formulation

In this section, the Relative localization problem between the pursuer and target robot is formulated.

A. Robot model

Consider a target robot and a pursuer robot in $\mathbb{R}^n$, where $n \in \{2, 3\}$. The kinematics models of the two robot are

$$\dot{p}_T = u_T, \qquad \dot{p}_O = u_O, \tag{2}$$

where $p_T, p_O \in \mathbb{R}^n$ are the positions, $u_T \in \mathbb{R}^n$ is the unknown target velocity input, and $u_O \in \mathbb{R}^n$ is the control input of the pursuer robot. Define the relative position, range, and bearing between the target and caputre robots as

$$r = p_T - p_O, \qquad \rho = \|r\|, \qquad b = \frac{r}{\|r\|}. \tag{3}$$

In this paper, we assume that the control input of the target is an unknown function of the bearing b, i.e. $u_T(b)$. We also assume that the control input u_T of the target is bounded.

Assumption 1. *The actual target input $u_T(t)$ is continuously differentiable and satisfies*

$$\|u_T(t)\| \leq \bar{u}_T, \qquad \|\dot{u}_T(t)\| \leq \bar{u}_{T,d},$$

where $\bar{u}_T$ and $\bar{u}_{T,d}$ are unknown positive constants.

Remark 1. *It is reasonable to model the target's evasive strategy as a function of the relative bearing, i.e., the line-of-sight (LOS) direction. From the perspective of target evasion, this modeling choice is motivated by two considerations. First, in the classical one-pursuer–one-evader differential game, where both agents have bounded speeds and are not subject to turning-rate constraints, the capture-time-maximizing strategy of the target is to move at its maximum speed directly away from the pursuer along the LOS direction. Second, relative bearing is one of the most readily available measurements between the target UAV and the pursuer and can often be obtained directly using onboard vision. In contrast, accurately obtaining relative range and velocity generally requires additional sensing capabilities or estimation procedures, particularly in non-cooperative sensing scenarios. Therefore, relative bearing provides a natural source of feedback for the target to formulate its evasive strategy.*

B. Localization and Control Object

Lemma 1. *For any continuous mapping $u_T : \Omega \to \mathbb{R}^n$, there exists an RBF neural network such that*

$$u_T(b) = \Phi(b)\theta^\star + \varepsilon_T(b), \tag{4}$$

where

$$\Phi(b) = I_n \otimes \phi^\top(b) \in \mathbb{R}^{n \times nq}, \qquad \theta^\star \in \mathbb{R}^{nq},$$

and

$$\phi(b) = \mathrm{col}\{\phi_1(b), \ldots, \phi_q(b)\} \in \mathbb{R}^q$$

is the known RBF basis vector, with

$$\phi_l(b) = \exp\left(-\frac{\|b - \mu_l\|^2}{2\sigma^2}\right), \qquad l = 1, \ldots, q.$$

Here, $\mu_l \in \mathbb{R}^n$ and $\sigma > 0$ are the center and width of the l-th Gaussian basis function, respectively. $\theta^\star \in \mathbb{R}^{nq}$ is the ideal weight vector, and $\varepsilon_T(b)$ is the approximation error satisfying

$$\|\varepsilon_T(b)\| \leq \bar{\varepsilon}_T, \tag{5}$$

where $\bar{\varepsilon}_T \in \mathbb{R}_+$ is an unknown finite constant. Since every Gaussian basis function is continuous and bounded on Ω, the regressor matrix $\Phi(b)$ is also uniformly bounded. Therefore, there exists a finite constant $\bar{\Phi} > 0$ such that

$$\mathrm{tr}\big[\Phi^\top(b)\Phi(b)\big] \leq \bar{\Phi}^2, \qquad \forall b \in \Omega. \tag{6}$$

III. Main Results

In this section, we first establish a relative-positioning model between the pursuer and the target and develop a data-driven estimator to estimate the target's evasion strategy online. Building on this result, we further design a target-relative-position estimator and a tracking controller for the pursuer.

A. Relative Localization Model Construction

The relative kinematics between the target and pursuer is

$$\dot{r} = \Phi(b)\theta^\star + \varepsilon_T - u_O. \tag{7}$$

When the bearing b is measurable, the real relative position r between the target and pursuer robot satisfies

$$P_b r = 0, \qquad P_b = I_3 - bb^\top.$$

Next, we first establish the observation equation between measurable information and unknown parameters. Define time instant $t_k = k\Delta T$, where $k = 0, 1, 2, \ldots$ is non-negative integer and $\Delta T > 0$ is the finite-window length. For each time instant t in $[t_{k-1}, t_k]$, integrating (7) over the closed interval $[t_{k-1}, t]$ gives

$$r(t) - r(t_{k-1}) = G_k(t)\theta^\star + E_k(t) - U_k(t). \tag{8}$$

where G_k, U_k and E_k are defined as

$$\begin{aligned} G_k(t) &= \int_{t_{k-1}}^{t} \Phi(b(\tau))d\tau, \\ U_k(t) &= \int_{t_{k-1}}^{t} u_O(\tau)d\tau, \\ E_k(t) &= \int_{t_{k-1}}^{t} \varepsilon_T(\tau)d\tau. \end{aligned} \tag{9}$$

Left-multiplying (8) by $P_b(t)$ yields

$$P_b(t)U_k(t) = P_b(t)G_k(t)\theta^\star + P_b(t)r(t_{k-1}) + P_b(t)E_k(t). \tag{10}$$

To remove the unknown $r(t_{k-1})$, define

$$a_k = \int_{t_{k-1}}^{t_k} b^\top(t_{k-1})P_b(\tau)b(t_{k-1})d\tau. \tag{11}$$

A window is called valid if $a_k > 0$. For f defined on $[t_{k-1}, t_k]$, define the finite-window projection operator, acting columnwise when f is matrix-valued, as

$$\begin{aligned} (\mathcal{M}_k f)(t) = &- P_b(t)b(t_{k-1})a_k^{-1}\int_{t_{k-1}}^{t_k} b^\top(t_{k-1})P_b(\tau)f(\tau)d\tau \\ &+ f(t) \quad t \in [t_{k-1}, t_k]. \end{aligned} \tag{12}$$

Since $r(t_{k-1}) = \rho(t_{k-1})b(t_{k-1})$, substituting $f(t) = P_b(t)r(t_{k-1})$ into (12) gives

$$\begin{aligned} &\mathcal{M}_k\left[P_b(\cdot)r(t_{k-1})\right](t) \\ =&P_b(t)b(t_{k-1})\rho(t_{k-1}) \\ &- P_b(t)b(t_{k-1})a_k^{-1}\int_{t_{k-1}}^{t_k} b^\top(t_{k-1})P_b(\tau)b(t_{k-1})\rho(t_{k-1})d\tau \\ =&P_b(t)b(t_{k-1})\rho(t_{k-1}) - P_b(t)b(t_{k-1})a_k^{-1}a_k\rho(t_{k-1}) = 0. \end{aligned} \tag{13}$$

Since $\mathcal{M}_k$ is linear and $\theta^\star$ is constant, applying $\mathcal{M}_k$ to both sides of (10) and using (13) gives that for every $t \in [t_{k-1}, t_k]$,

$$\mathcal{M}_k[P_bU_k](t) = \mathcal{M}_k[P_bG_k](t)\theta^\star + \mathcal{M}_k[P_bE_k](t). \tag{14}$$

Premultiplying (14) by $(\mathcal{M}_k[P_bG_k](t))^\top$ yields the pointwise normal equation

$$\begin{aligned} &(\mathcal{M}_k[P_bG_k](t))^\top\mathcal{M}_k[P_bU_k](t) \\ =&(\mathcal{M}_k[P_bG_k](t))^\top\mathcal{M}_k[P_bG_k](t)\theta^\star \\ &+ (\mathcal{M}_k[P_bG_k](t))^\top\mathcal{M}_k[P_bE_k](t). \end{aligned} \tag{15}$$

Integrating both sides of (15) over $[t_{k-1}, t_k]$ gives

$$\begin{aligned} &\int_{t_{k-1}}^{t_k} \left(\mathcal{M}_k[P_bG_k]\right)^\top(\tau)\mathcal{M}_k[P_bU_k](\tau)d\tau \\ =&\left[\int_{t_{k-1}}^{t_k} \left(\mathcal{M}_k[P_bG_k]\right)^\top(\tau)\mathcal{M}_k[P_bG_k](\tau)d\tau\right]\theta^\star \\ &+ \int_{t_{k-1}}^{t_k} \left(\mathcal{M}_k[P_bG_k]\right)^\top(\tau)\mathcal{M}_k[P_bE_k](\tau)d\tau. \end{aligned} \tag{16}$$

Define the finite-window information matrix and vector by

$$\begin{aligned} W_k &= \int_{t_{k-1}}^{t_k} \left(\mathcal{M}_k[P_bG_k]\right)^\top(\tau)\left(\mathcal{M}_k[P_bG_k]\right)(\tau)d\tau, \\ Z_k &= \int_{t_{k-1}}^{t_k} \left(\mathcal{M}_k[P_bG_k]\right)^\top(\tau)\left(\mathcal{M}_k[P_bU_k]\right)(\tau)d\tau. \\ \Delta_k &= \int_{t_{k-1}}^{t_k} \left(\mathcal{M}_k[P_bG_k]\right)^\top(\tau)\left(\mathcal{M}_k[P_bE_k]\right)(\tau)d\tau. \end{aligned}$$

Therefore,

$$Z_k = W_k\theta^\star + \Delta_k,$$

A finite history stack $\mathcal{S}_h$ with at most N_h retained windows is maintained. The retained information matrix and vector are

$$W_h = \sum_{k\in\mathcal{S}_h} W_k, \qquad Z_h = \sum_{k\in\mathcal{S}_h} Z_k.$$

The retained finite-window LRE is

$$Z_h = W_h\theta^\star + \Delta_h, \qquad \Delta_h = \sum_{i\in\mathcal{S}_h} \Delta_i. \tag{17}$$

The completed-window LRE above preserves informative historical excitation, but it is available only at the window-ending instants. To also exploit the data being generated in the current window, consider $t \in [t_k, t_{k+1})$, where t now denotes the actual online current time, define scalar $a_{c,k}$ as

$$a_{c,k}(t) = \int_{t_k}^{t} b^\top(t_k)P_b(\tau)b(t_k)d\tau.$$

Choose $\underline{a}_c > 0$, and define

$$\chi_{c,k}(t) = \begin{cases} 1, & a_{c,k}(t) \geq \underline{a}_c, \\ 0, & a_{c,k}(t) < \underline{a}_c. \end{cases}$$

For $\chi_{c,k}(t) = 0$, set $W_{c,k}(t) = 0$, $Z_{c,k}(t) = 0$. For $\chi_{c,k}(t) = 1$ and time instant $\sigma \in [t_k, t]$, define the causal current-window operator

$$\begin{aligned} (\mathcal{M}_{c,k}[f])(\sigma) = f(\sigma) - P_b(\sigma)b(t_k)a_{c,k}^{-1}(t) \\ \times \int_{t_k}^{t} b^\top(t_k)P_b(\tau)f(\tau)d\tau. \end{aligned} \tag{18}$$

Using the partial functions $G_{k+1}(\tau)$, $U_{k+1}(\tau)$, and $E_{k+1}(\tau)$ from (9), whose lower integration limit is t_k, define

$$\begin{aligned} W_{c,k}(t) &= \int_{t_k}^{t} \left(\mathcal{M}_{c,k}[P_bG_{k+1}]\right)^\top(\tau) \times \left(\mathcal{M}_{c,k}[P_bG_{k+1}]\right)(\tau)d\tau, \\ Z_{c,k}(t) &= \int_{t_k}^{t} \left(\mathcal{M}_{c,k}[P_bG_{k+1}]\right)^\top(\tau) \times \left(\mathcal{M}_{c,k}[P_bU_{k+1}]\right)(\tau)d\tau, \\ \Delta_{c,k}(t) &= \int_{t_k}^{t} \left(\mathcal{M}_{c,k}[P_bG_{k+1}]\right)^\top(\tau) \times \left(\mathcal{M}_{c,k}[P_bE_{k+1}]\right)(\tau)d\tau. \end{aligned} \tag{19}$$

These quantities satisfy the causal current-window LRE

$$Z_{c,k}(t) = W_{c,k}(t)\theta^\star + \Delta_{c,k}(t), \tag{20}$$

where $\Delta_{c,k}(t)$ is the approximation residual accumulated only over the current partial window. At $t = t_{k+1}$, the causal operator $\mathcal{M}_{c,k}$ coincides with the completed-window operator $\mathcal{M}_{k+1}$. The pair is then finalized as (W_{k+1}, Z_{k+1}), considered by the history-stack replacement rule, and the current statistics are reset to zero.

B. Data driven parameter estimator

The history stack is updated only at the discrete window-ending instants t_k. Starting from $\mathcal{S}_h = \emptyset$, once a new valid pair (W_k, Z_k) is generated, it is directly stored if the stack is not full. If the stack already contains N_h pairs, the new pair is compared with each stored pair as a possible replacement. Among all possible single-pair replacements, retain the candidate whose aggregated information matrix has the largest vector $\lambda^{\uparrow}$. Perform the replacement only when that vector is strictly larger than the vector of the current aggregate; otherwise leave the stack unchanged. If no valid pair is generated at t_k, the stack is kept unchanged. In all cases,

$$|\mathcal{S}_h| \le N_h.$$

The retained relation (17) is a linear regression in $\theta^\star$. If W_h is uniformly positive definite, one could directly multiply it by W_h^{-1}. However, is either undefined or strongly amplifies Δ_h when the retained data do not excite all parameter directions. To estimate the control strategy parameters of non cooperative targets as much as possible even when the incentive information is insufficient, we propose the following spectral decomposition based method. At each history-stack update instant t_k, let the eigendecomposition of W_h be given by $W_h = Q_k \Lambda_k Q_k^\top = \sum_{\ell=1}^{nq} \lambda_{\ell,k} E_{\ell,k}$, where $\Lambda_k = \operatorname{diag} \lambda_{1,k}, \dots, \lambda_{nq,k}$ with $\lambda_{\ell,k} \ge 0$, $Q_k = [q_{1,k}, \dots, q_{nq,k}]$ is an orthogonal matrix, and $E_{\ell,k} = q_{\ell,k} q_{\ell,k}^\top$ denotes the rank-one orthogonal projector onto the eigendirection associated with $q_{\ell,k}$. Hence

$$\begin{aligned} E_{\ell,k}^2 = E_{\ell,k}, \qquad & E_{\ell,k} E_{j,k} = 0 \quad (\ell \ne j), \\ E_{\ell,k} W_h = W_h E_{\ell,k}, \quad & W_h E_{\ell,k} = \lambda_{\ell,k} E_{\ell,k}. \end{aligned}$$

Premultiplying the retained LRE (17) by $E_{\ell,k}$ gives the spectral parameter observation

$$E_{\ell,k} Z_h = \lambda_{\ell,k} E_{\ell,k} \theta^\star + E_{\ell,k} \Delta_h, \qquad \ell = 1, \dots, nq. \quad (21)$$

Thus $\lambda_{\ell,k}$ is the data richness for the projected parameter component $E_{\ell,k}\theta^\star$. If $\lambda_{\ell,k}$ is large, then $E_{\ell,k} Z_h$ gives a reliable linear observation of $E_{\ell,k}\theta^\star$; if $\lambda_{\ell,k}$ is close to zero, this component is weakly excited and should not be inverted. Formally, whenever $\lambda_{\ell,k} > 0$,

$$E_{\ell,k}\theta^\star = \frac{1}{\lambda_{\ell,k}} E_{\ell,k} Z_h - \frac{1}{\lambda_{\ell,k}} E_{\ell,k} \Delta_h.$$

This relation motivates estimating only the sufficiently excited spectral directions rather than requiring the full matrix W_h to be nonsingular. Choose an excitation threshold $\gamma_\theta > 0$, and define the active index set

$$\mathcal{I}_{\gamma,k} = \{\ell : \lambda_{\ell,k} \ge \gamma_\theta\}.$$

The active-subspace projector and the truncated inverse are defined as

$$\Pi_{\gamma,k} = \sum_{\ell \in \mathcal{I}_{\gamma,k}} E_{\ell,k}, \qquad W_{\gamma,k}^\dagger = \sum_{\ell \in \mathcal{I}_{\gamma,k}} \frac{1}{\lambda_{\ell,k}} E_{\ell,k}.$$

If $\mathcal{I}_{\gamma,k} = \emptyset$, then $\Pi_{\gamma,k} = 0$ and $W_{\gamma,k}^\dagger = 0$. By construction,

$$W_{\gamma,k}^\dagger W_h = \Pi_{\gamma,k}.$$

Multiplying (17) by $W_{\gamma,k}^\dagger$ yields the active-subspace relation

$$W_{\gamma,k}^\dagger Z_h = \Pi_{\gamma,k}\theta^\star + W_{\gamma,k}^\dagger \Delta_h. \quad (22)$$

Therefore $W_{\gamma,k}^\dagger Z_h$ is not intended to estimate the full vector $\theta^\star$. It is the data-driven estimate of the currently identifiable projection $\Pi_{\gamma,k}\theta^\star$, with an error bounded by the projected residual $W_{\gamma,k}^\dagger \Delta_h$. Design

$$\hat{\theta}_{\gamma,k} = W_{\gamma,k}^\dagger Z_h. \quad (23)$$

It follows immediately that

$$\hat{\theta}_{\gamma,k} - \Pi_{\gamma,k}\theta^\star = W_{\gamma,k}^\dagger \Delta_h.$$

When all the parameter directions are sufficiently excited, $\Pi_{\gamma,k} = I_{nq}$, and (23) reduces to the standard full-parameter batch estimate. Otherwise, it provides only the reliably identifiable part of the parameter vector. For $t \in [t_k, t_{k+1})$, design the continuous-time parameter update law as

$$\begin{aligned} \dot{\hat{\theta}}(t) = {} & k_c \chi_{c,k}(t) \big[Z_{c,k}(t) - W_{c,k}(t)\hat{\theta}(t) \big] \\ & + k_h \big[\hat{\theta}_{\gamma,k} - \Pi_{\gamma,k}\hat{\theta}(t) \big] - \sigma_\theta \big(I_{nq} - \Pi_{\gamma,k} \big) \hat{\theta}(t), \end{aligned} \quad (24)$$

where $k_c > 0$, $k_h > 0$, and $\sigma_\theta > 0$. Define the parameter estimation error as $\tilde{\theta} = \theta^\star - \hat{\theta}$. By (20) and (22), the two data-driven corrections satisfy

$$\begin{aligned} Z_{c,k} - W_{c,k}\hat{\theta} &= W_{c,k}\tilde{\theta} + \Delta_{c,k}, \\ \hat{\theta}_{\gamma,k} - \Pi_{\gamma,k}\hat{\theta} &= \Pi_{\gamma,k}\tilde{\theta} + W_{\gamma,k}^\dagger \Delta_h. \end{aligned}$$

Therefore, the first term of (24) uses the data being generated in the current partial window. The second term uses the retained history only in the reliably excited parameter subspace. The last term does not identify the inactive parameter component, it only suppresses drift in directions for which the retained history provides insufficient information. Whenever new completed windows are admitted to the history stack, the spectrum of W_h, the active projector $\Pi_{\gamma,k}$, and the history-based target $\hat{\theta}_{\gamma,k}$ are recomputed. The stability of the estimator (24) can be explained by the following Lemma

Lemma 2. *Define* $c_\theta = \min\{k_h, \sigma_\theta\}$ *and* $\bar{d}_\theta = \Delta T^3 \bar{\Phi} \bar{\varepsilon}_T (k_c + \frac{k_h N_h}{\gamma_\theta}) + \sigma_\theta \|\theta^\star\|$, *then for all* $t \ge 0$,

$$\left\| \tilde{\theta}(t) \right\| \le e^{-c_\theta t} \left\| \tilde{\theta}(0) \right\| + \frac{\bar{d}_\theta}{c_\theta} \big(1 - e^{-c_\theta t} \big). \quad (25)$$

Consequently, define $\bar{\tilde{\theta}} = \max\left\{ \left\| \tilde{\theta}(0) \right\|, \frac{\bar{d}_\theta}{c_\theta} \right\}$, *then*

$$\left\| \tilde{\theta}(t) \right\| \le \bar{\tilde{\theta}}, \qquad \left\| \hat{\theta}(t) \right\| \le \bar{\theta} = \|\theta^\star\| + \bar{\tilde{\theta}}, \qquad \forall t \ge 0. \quad (26)$$

Proof. The history stack $\mathcal{S}_h$ keeps at most N_h windows. Consider valid completed window $[t_{k-1}, t_k]$. For every $t \in [t_{k-1}, t_k]$, Lemma 1 gives

$$\begin{aligned} \operatorname{tr}\big(G_k^\top(t) G_k(t) \big) &\le (t - t_{k-1})^2 \bar{\Phi}^2, \\ \|E_k(t)\|^2 &\le (t - t_{k-1})^2 \bar{\varepsilon}_T^2. \end{aligned} \quad (27)$$

Let $v_k(t) = P_b(t) b(t_{k-1})$ and $C_{k,f} = \int_{t_{k-1}}^{t_k} v_k^\top(s) f(s) ds$, since $P_b^\top(t) = P_b(t)$ and $P_b^2(t) = P_b(t)$, then it has

$$a_k = \int_{t_{k-1}}^{t_k} \|v_k(\tau)\|^2 \, d\tau,$$

and (12) becomes

$$(\mathcal{M}_k f)(t) = f(t) - \frac{1}{a_k} v_k(t) C_{k,f}.$$

Direct expansion using $\mathrm{tr}(A^\top B)$ gives

$$\begin{aligned}
&\int_{t_{k-1}}^{t_k} \mathrm{tr}\big[(\mathcal{M}_k f)^\top(\tau)(\mathcal{M}_k f)(\tau)\big]\, d\tau \\
&= \int_{t_{k-1}}^{t_k} \mathrm{tr}\big[f^\top(\tau) f(\tau)\big]\, d\tau \\
&\quad - \frac{2}{a_k}\,\mathrm{tr}\left[\left(\int_{t_{k-1}}^{t_k} f^\top(\tau) v_k(\tau) d\tau\right) C_{k,f}\right] \\
&\quad + \frac{1}{a_k^2}\int_{t_{k-1}}^{t_k} \mathrm{tr}\big[C_{k,f}^\top v_k^\top(\tau) v_k(\tau) C_{k,f}\big]\, d\tau \\
&= \int_{t_{k-1}}^{t_k} \mathrm{tr}\big[f^\top(\tau) f(\tau)\big]\, d\tau - \frac{\|C_{k,f}\|^2}{a_k} \\
&\le \int_{t_{k-1}}^{t_k} \mathrm{tr}\big[f^\top(\tau) f(\tau)\big]\, d\tau. \qquad (28)
\end{aligned}$$

Applying (28) to $f = P_b G_k$ and using $P_b^\top P_b = P_b \preceq I_n$ and (27) yield

$$\begin{aligned}
&\int_{t_{k-1}}^{t_k} \mathrm{tr}\Big[\left(\mathcal{M}_k[P_b G_k](\tau)\right)^\top \mathcal{M}_k[P_b G_k](\tau)\Big]\, d\tau \\
&\quad \le \int_{t_{k-1}}^{t_k} \mathrm{tr}\big[G_k^\top(\tau) G_k(\tau)\big]\, d\tau \\
&\quad \le \frac{\Delta T^3}{3}\bar{\Phi}^2 \le \Delta T^3 \bar{\Phi}^2. \qquad (29)
\end{aligned}$$

Similarly, from (27),

$$\int_{t_{k-1}}^{t_k} \left\|\mathcal{M}_k[P_b E_k](\tau)\right\|^2 d\tau \le \frac{\Delta T^3}{3}\bar{\varepsilon}_T^2 \le \Delta T^3 \bar{\varepsilon}_T^2. \qquad (30)$$

Using $\|A^\top e\|^2 \le \mathrm{tr}(A^\top A)\|e\|^2$ and the Cauchy–Schwarz inequality, the definition of Δ_k gives

$$\begin{aligned}
\|\Delta_k\| &= \left\| \int_{t_{k-1}}^{t_k} \left(\mathcal{M}_k[P_b G_k](\tau)\right)^T \mathcal{M}_k[P_b E_k](\tau) d\tau \right\| \\
&\le \left(\int_{t_{k-1}}^{t_k} \mathrm{tr}\big[\left(\mathcal{M}_k[P_b G_k](\tau)\right)^T \mathcal{M}_k[P_b G_k](\tau)\big]\, d\tau \right)^{1/2} \\
&\quad \times \left(\int_{t_{k-1}}^{t_k} \left\|\mathcal{M}_k[P_b E_k](\tau)\right\|^2 d\tau \right)^{1/2} \\
&\le \Delta T^3 \bar{\Phi} \bar{\varepsilon}_T. \qquad (31)
\end{aligned}$$

Sine the history stack retains at most N_h completed windows,

$$\|\Delta_h\| \le \sum_{k \in \mathcal{S}_h} \|\Delta_k\| \le N_h \Delta T^3 \bar{\Phi} \bar{\varepsilon}_T. \qquad (32)$$

For $t \in [t_k, t_{k+1})$, if $\chi_{c,k}(t) = 0$, then $\Delta_{c,k}(t) = 0$ by definition. If $\chi_{c,k}(t) = 1$, the same calculation over the available interval $[t_k, t]$ yields

$$\|\Delta_{c,k}(t)\| \le (t - t_k)^3 \bar{\Phi} \bar{\varepsilon}_T \le \Delta T^3 \bar{\Phi} \bar{\varepsilon}_T. \qquad (33)$$

Since $\Pi_{\gamma,k}$ is an orthogonal projector, one has

$$\widetilde{\theta}^\top \Pi_{\gamma,k} \widetilde{\theta} = \left\|\Pi_{\gamma,k}\widetilde{\theta}\right\|^2,$$
$$\widetilde{\theta}^\top (I_{nq} - \Pi_{\gamma,k}) \widetilde{\theta} = \left\|(I_{nq} - \Pi_{\gamma,k})\widetilde{\theta}\right\|^2,$$

and the two projected components are orthogonal. Therefore,

$$\left\|\widetilde{\theta}\right\|^2 = \left\|\Pi_{\gamma,k}\widetilde{\theta}\right\|^2 + \left\|(I_{nq} - \Pi_{\gamma,k})\widetilde{\theta}\right\|^2. \qquad (34)$$

Differentiating V_θ and substituting (24) give

$$\begin{aligned}
\dot{V}_\theta = &- k_c \chi_{c,k}(t) \widetilde{\theta}^\top W_{c,k}(t) \widetilde{\theta} \\
&- k_h \left\|\Pi_{\gamma,k}\widetilde{\theta}\right\|^2 - \sigma_\theta \left\|(I_{nq} - \Pi_{\gamma,k})\widetilde{\theta}\right\|^2 \\
&- k_c \chi_{c,k}(t) \widetilde{\theta}^\top \Delta_{c,k}(t) - k_h \widetilde{\theta}^\top W_{\gamma,k}^\dagger \Delta_h \\
&+ \sigma_\theta \widetilde{\theta}^\top (I_{nq} - \Pi_{\gamma,k}) \theta^\star. \qquad (35)
\end{aligned}$$

Recall that $c_\theta = \min\{k_h, \sigma_\theta\}$, using (34), the two projected decay terms satisfy

$$-k_h \left\|\Pi_{\gamma,k}\widetilde{\theta}\right\|^2 - \sigma_\theta \left\|(I_{nq} - \Pi_{\gamma,k})\widetilde{\theta}\right\|^2 \le -c_\theta \left\|\widetilde{\theta}\right\|^2.$$

Since $W_{c,k}(t) \succeq 0$, applying the Cauchy–Schwarz inequality to the remaining terms in (35) yields

$$\begin{aligned}
\dot{V}_\theta \le &- k_c \chi_{c,k}(t) \widetilde{\theta}^\top W_{c,k}(t) \widetilde{\theta} - c_\theta \left\|\widetilde{\theta}\right\|^2 \qquad (36) \\
&+ \left\|\widetilde{\theta}\right\| \Big(k_c \left\|\Delta_{c,k}(t)\right\| + k_h \left\|W_{\gamma,k}^\dagger \Delta_h\right\| + \sigma_\theta \left\|\theta^\star\right\| \Big).
\end{aligned}$$

Here, we have used $W_{c,k}(t) \succeq 0$, $\chi_{c,k}(t) \le 1$ and $\left\|(I_{nq} - \Pi_{\gamma,k})\widetilde{\theta}\right\| \le \left\|\widetilde{\theta}\right\|$. Furthermore, the definition of the truncated inverse gives

$$\left\|W_{\gamma,k}^\dagger\right\| \le \frac{1}{\gamma_\theta}. \qquad (37)$$

Combining this inequality with (32), (33) and (37) gives,

$$\begin{aligned}
\dot{V}_\theta &\le - k_c \chi_{c,k}(t) \widetilde{\theta}^\top W_{c,k}(t) \widetilde{\theta} - c_\theta \left\|\widetilde{\theta}\right\|^2 + \bar{d}_\theta \left\|\widetilde{\theta}\right\| \\
&\le - c_\theta \left\|\widetilde{\theta}\right\|^2 + \bar{d}_\theta \left\|\widetilde{\theta}\right\|.
\end{aligned}$$

Since $V_\theta = \frac{1}{2}\left\|\widetilde{\theta}\right\|^2$, it follows that

$$\dot{V}_\theta \le -2c_\theta V_\theta + \sqrt{2}\bar{d}_\theta \sqrt{V_\theta}. \qquad (38)$$

Consider the scalar comparison system

$$\dot{\overline{V}}_\theta = -2c_\theta \overline{V}_\theta + \sqrt{2}\bar{d}_\theta \sqrt{\overline{V}_\theta}, \qquad \overline{V}_\theta(0) = V_\theta(0). \qquad (39)$$

Its maximal nonnegative solution can be constructed by letting $q(t)$ solve

$$\dot{q} = -c_\theta q + \bar{d}_\theta, \qquad q(0) = \sqrt{2V_\theta(0)}. \qquad (40)$$

Hence

$$q(t) = e^{-c_\theta t}\sqrt{2V_\theta(0)} + \frac{\bar{d}_\theta}{c_\theta}\left(1 - e^{-c_\theta t}\right). \qquad (41)$$

Since $q(t) \ge 0$, the function

$$\overline{V}_\theta(t) = \frac{1}{2} q^2(t) \qquad (42)$$

satisfies (39). Thus,

$$\overline{V}_\theta(t) = \frac{1}{2}\left[e^{-c_\theta t}\sqrt{2V_\theta(0)} + \frac{\bar{d}_\theta}{c_\theta}\left(1-e^{-c_\theta t}\right)\right]^2. \tag{43}$$

Applying the scalar comparison principle to (38) and (39) gives

$$V_\theta(t) \le \overline{V}_\theta(t), \qquad t \ge 0. \tag{44}$$

Consequently,

$$\begin{aligned}\left\|\widetilde{\theta}(t)\right\| &\le \sqrt{2\overline{V}_\theta(t)} \\ &= e^{-c_\theta t}\left\|\widetilde{\theta}(0)\right\| + \frac{\bar{d}_\theta}{c_\theta}\left(1-e^{-c_\theta t}\right),\end{aligned} \tag{45}$$

which proves (25). The uniform bound (26) follows immediately from this estimate and $\left\|\widehat{\theta}\right\| \le \|\theta^\star\| + \left\|\widetilde{\theta}\right\|$. □

The RBF-based target-input estimate is passed through the first-order filter

$$\dot{\widehat{u}}_T = a_f\left(\Phi(b)\widehat{\theta} - \widehat{u}_T\right), \qquad a_f > 0. \tag{46}$$

The target-input estimate error is defined as

$$\delta_v = u_T - \widehat{u}_T \tag{47}$$

Remark 2. *Unlike existing results that model a noncooperative target as stationary or moving with constant velocity or acceleration, in this paper, we considers a target that actively executes an evasion policy unknown to the pursuer. The proposed strategy estimator* (46) *is designed to make full use of both historical and current measurement information to estimate the target's escape strategy, thereby predicting its future control input to some extent and providing a feedforward estimate that complements the feedback estimation in the subsequent relative-position estimator.*

C. Relative Localization Estimator Design

Based on the estimated target control strategy, we develop an extended-state observer (ESO) to estimate the relative position between the pursuer and the target, thereby achieving relative localization. Define the augmented state as

$$\begin{aligned}\xi_o &= \mathrm{col}\{r, \delta_v\} \in \mathbb{R}^{2n}, \\ E_o &= \mathrm{col}\{0_n, I_n\}.\end{aligned}$$

Then

$$\dot{r} = \widehat{u}_T - u_O + \delta_v, \tag{48}$$

and the augmented dynamics can be written as

$$\dot{\xi}_o = A_o\xi_o + g_o + E_o w_o,$$

where

$$\begin{aligned}g_o &= \mathrm{col}\{\widehat{u}_T - u_O, 0_n\}, \\ w_o &= \dot{\delta}_v = \dot{u}_T - a_f\left(\Phi(b)\widehat{\theta} - \widehat{u}_T\right).\end{aligned} \tag{49}$$

The bearing constraint provides the virtual output $y_o = C_o(t)\xi_o = 0$. Consider the Riccati gain generator

$$\begin{aligned}&\dot{P}_o = A_oP_o + P_oA_o^\top + \lambda_sP_o - P_oC_o^\top(t)R_o^{-1}C_o(t)P_o, \\ &P_o(0) = P_o^\top(0) > 0,\end{aligned} \tag{50}$$

where $P_o(t) \in \mathbb{R}^{2n\times 2n}$, $R_o = R_o^\top > 0 \in \mathbb{R}^{n\times n}$, and $\lambda_s > 0$. The relative-localization estimator is

$$\dot{\widehat{\xi}}_o = A_o\widehat{\xi}_o + g_o + L_o(t)\left(y_o - C_o(t)\widehat{\xi}_o\right), \tag{51}$$

with $L_o(t) = P_o(t)C_o^\top(t)R_o^{-1}$. Let $\widehat{\xi}_o = \mathrm{col}\{\widehat{r}, \widehat{\delta}_v\}$. Define $\widetilde{\xi}_o = \xi_o - \widehat{\xi}_o = \mathrm{col}\{\widetilde{r}, \widetilde{\delta}_v\}$. Since $y_o = C_o\xi_o = 0$, the localization error satisfies

$$\dot{\widetilde{\xi}}_o = (A_o - L_oC_o)\widetilde{\xi}_o + E_ow_o. \tag{52}$$

To guarantee the convergence of the relative positioning estimator (51), the following assumptions are required.

Assumption 2. *Let* $A_o = \begin{bmatrix}0_n & I_n \\ 0_n & 0_n\end{bmatrix}$ *and* $C_o(t) = \begin{bmatrix}P_b(t) & 0_n\end{bmatrix}$. *There exist constants* $T_o > 0$ *and* $\alpha_o > 0$ *such that, for all* $t \ge T_o$,

$$\begin{aligned}\mathcal{G}_o(t) = &\int_{t-T_o}^{t}\Psi_o^{-\top}(t,\tau)C_o^\top(\tau)C_o(\tau)\Psi_o^{-1}(t,\tau)d\tau \\ &\ge \alpha_oI_{2n},\end{aligned} \tag{53}$$

where $\Psi_o(t,\tau) = e^{A_o(t-\tau)} = \begin{bmatrix}I_n & (t-\tau)I_n \\ 0_n & I_n\end{bmatrix}$. *Equivalently* $\mathcal{G}_o(t) = \int_{t-T_o}^{t}\begin{bmatrix}P_b(\tau) & -(t-\tau)P_b(\tau) \\ -(t-\tau)P_b(\tau) & (t-\tau)^2P_b(\tau)\end{bmatrix}d\tau$. *This is the uniform complete observability condition for the bearing-only pair* $(A_o, C_o(t))$.

Remark 3. *In the relative localization estimator* (51)*, the feedforward term* g_o *is constructed using the estimate of the target control strategy. As indicated by the positioning-error dynamics, unlike approaches that treat the entire target control input* u_T *as an unknown disturbance, the proposed method treats only its estimation error,* $\delta_v = u_T - \hat{u}_T$*, as a disturbance. This allows the available measurement information to be better utilized and can potentially improve the relative-positioning accuracy. The effectiveness of this feedforward mechanism will be further evaluated through ablation studies in the simulation section.*

Remark 4. *The essence of Assumption 53 is that the relative bearing between the pursuer and the target should not remain unchanged for an extended period. This condition ensures that the bearing measurements provide sufficiently rich excitation for relative localization and is commonly imposed in bearing-only relative-localization studies. It can generally be satisfied by appropriately designing the desired relative position between the pursuer and the target, with spiral tracking being a common choice.*

Before proving the stability of the relative localization estimator (51), we first provide the following Lemma

Lemma 3. *Suppose Assumption 2 holds and* $P_o(0) > 0$*. Then there exist constants* $p_m > 0$ *and* $p_M > 0$ *such that*

$$p_mI_{2n} \le P_o(t) \le p_MI_{2n}, \qquad \forall t \ge 0. \tag{54}$$

Proof. Define $A_{\lambda,o} = A_o + \frac{\lambda_s}{2}I_{2n}$ and $\Psi_{\lambda,o}(t,\tau) = e^{A_{\lambda,o}(t-\tau)}$, then (50) becomes

$$\dot{P}_o = A_{\lambda,o}P_o + P_oA_{\lambda,o}^\top - P_oC_o^\top R_o^{-1}C_oP_o. \tag{55}$$

Consider the linear matrix equation $\dot{\Omega}_o = -\Omega_o A_{\lambda,o} - A_{\lambda,o}^\top \Omega_o + C_o^\top R_o^{-1} C_o$ and $\Omega_o(0) = P_o^{-1}(0)$. Its solution is

$$\Omega_o(t) = \Psi_{\lambda,o}^{-\top}(t,0) P_o^{-1}(0) \Psi_{\lambda,o}^{-1}(t,0) + \int_0^t \Psi_{\lambda,o}^{-\top}(t,\tau) C_o^\top(\tau) R_o^{-1} C_o(\tau) \Psi_{\lambda,o}^{-1}(t,\tau) d\tau. \tag{56}$$

Since the first term is positive definite and the second one is positive semidefinite, $\Omega_o(t) > 0$ for all $t \geq 0$. Moreover, direct differentiation of Ω_o^{-1} shows that Ω_o^{-1} satisfies (55) with the same initial condition. Hence $P_o(t) = \Omega_o^{-1}(t)$.

We first derive a lower bound for $\Omega_o(t)$. Since$\Psi_{\lambda,o}^{-1}(t,\tau) = e^{-\frac{\lambda_s}{2}(t-\tau)} \Psi_o^{-1}(t,\tau)$, for $t \geq T_o$, retaining only the most recent interval $[t - T_o, t]$ in (56) gives

$$\Omega_o(t) \geq e^{-\lambda_s T_o} \lambda_{\min}(R_o^{-1}) \mathcal{G}_o(t) \geq \rho_{\lambda,1} I_{2n},$$

where$\rho_{\lambda,1} = e^{-\lambda_s T_o} \lambda_{\min}(R_o^{-1}) \alpha_o > 0$. For $t \in [0, T_o]$, define

$$\rho_{\lambda,0} = \min_{t \in [0,T_o]} \lambda_{\min}\Big(\Psi_{\lambda,o}^{-\top}(t,0) P_o^{-1}(0) \Psi_{\lambda,o}^{-1}(t,0) \Big) > 0.$$

Therefore,

$$\Omega_o(t) \geq \underline{\rho}_\lambda I_{2n}, \qquad \underline{\rho}_\lambda = \min\{\rho_{\lambda,0}, \rho_{\lambda,1}\} > 0,$$

and hence

$$P_o(t) \leq p_M I_{2n}, \qquad p_M = \underline{\rho}_\lambda^{-1}. \tag{57}$$

We next derive an upper bound for $\Omega_o(t)$. Let $s = t - \tau \geq 0$. Then

$$\left\| \Psi_{\lambda,o}^{-1}(t,\tau) \right\|_2 \leq (1+s) e^{-\frac{\lambda_s}{2} s}.$$

For any $\alpha_\lambda \in (0, \lambda_s/2)$, let $\delta_\lambda = \frac{\lambda_s}{2} - \alpha_\lambda > 0$. Since the continuous function $(1+s)e^{-\delta_\lambda s}$ is bounded on $s \geq 0$, there exists a finite constant $\kappa_\lambda > 0$ such that $(1+s)e^{-\delta_\lambda s} \leq \kappa_\lambda, \quad \forall s \geq 0$. Therefore,

$$\left\| \Psi_{\lambda,o}^{-1}(t,\tau) \right\|_2 \leq \kappa_\lambda e^{-\alpha_\lambda (t-\tau)}, \qquad t \geq \tau.$$

In addition, since $\|C_o(t)\|_2 \leq 1$,

$$C_o^\top(t) R_o^{-1} C_o(t) \leq \lambda_{\max}(R_o^{-1}) I_{2n}.$$

Substituting these bounds into (56) yields

$$\Omega_o(t) \leq \overline{\rho}_\lambda I_{2n},$$

where $\overline{\rho}_\lambda = \kappa_\lambda^2 \left[\lambda_{\max}\big(P_o^{-1}(0)\big) + \frac{\lambda_{\max}(R_o^{-1})}{2\alpha_\lambda} \right] < \infty$. Consequently,

$$P_o(t) \geq p_m I_{2n}, \qquad p_m = \overline{\rho}_\lambda^{-1} > 0. \tag{58}$$

Combining (57) and (58) proves (54). □

The stability of the proposed relative localization estimator can be described in the following Lemma.

Lemma 4. *Suppose that Assumptions 1 and 2 and the conditions of Lemma 1 hold. Define*

$$\begin{aligned} \bar{u}_{T,e} &= \max\left\{ \|\widehat{u}_T(0)\|_2, \bar{\Phi}\bar{\theta} \right\}, \\ \bar{w}_o &= \bar{u}_{T,d} + a_f \left(\bar{\Phi}\bar{\theta} + \bar{u}_{T,e} \right), \\ \bar{\xi}_o &= \sqrt{\frac{p_M}{p_m}} \left\| \widetilde{\xi}_o(0) \right\|_2 + \frac{2}{\lambda_s} \sqrt{\frac{p_M}{p_m}}\, \bar{w}_o. \end{aligned}$$

Then

$$\left\| \widetilde{\xi}_o(t) \right\| \leq \bar{\xi}_o, \qquad \forall t \geq 0.$$

Proof. Consider

$$V_o = \widetilde{\xi}_o^\top P_o^{-1} \widetilde{\xi}_o. \tag{59}$$

It follows from (50) that

$$\frac{d}{dt} P_o^{-1} = -P_o^{-1} A_o - A_o^\top P_o^{-1} - \lambda_s P_o^{-1} + C_o^\top R_o^{-1} C_o.$$

Using $P_o^{-1} L_o C_o = C_o^\top R_o^{-1} C_o$, differentiation of (59) yields

$$\begin{aligned} \dot{V}_o =& -\lambda_s V_o - \widetilde{\xi}_o^\top C_o^\top R_o^{-1} C_o \widetilde{\xi}_o + 2\widetilde{\xi}_o^\top P_o^{-1} E_o w_o \\ \leq& -\lambda_s V_o + 2\widetilde{\xi}_o^\top P_o^{-1} E_o w_o. \end{aligned}$$

By Lemma 3 and Young's inequality,

$$\begin{aligned} 2\widetilde{\xi}_o^\top P_o^{-1} E_o w_o &\leq \frac{\lambda_s}{2} V_o + \frac{2}{\lambda_s} w_o^\top E_o^\top P_o^{-1} E_o w_o \\ &\leq \frac{\lambda_s}{2} V_o + \frac{2}{\lambda_s p_m} \|w_o\|_2^2. \end{aligned}$$

Therefore,

$$\dot{V}_o \leq -\frac{\lambda_s}{2} V_o + \frac{2}{\lambda_s p_m} \|w_o\|_2^2. \tag{60}$$

Lemma 2 gives $\|\widehat{\theta}(t)\|_2 \leq \bar{\theta}$. Hence, the solution of (46) satisfies

$$\|\widehat{u}_T(t)\|_2 \leq \bar{u}_{T,e}.$$

Using Assumption 1 in (49) gives

$$\|w_o(t)\|_2 \leq \bar{u}_{T,d} + a_f \left(\bar{\Phi}\bar{\theta} + \bar{u}_{T,e} \right) = \bar{w}_o. \tag{61}$$

Substituting (61) into (60) and applying the comparison Lemma give

$$V_o(t) \leq e^{-\frac{\lambda_s}{2} t} V_o(0) + \frac{4}{\lambda_s^2 p_m} \bar{w}_o^2. \tag{62}$$

Since $P_o(t) \leq p_M I_{2n}$ and $P_o^{-1}(0) \leq p_m^{-1} I_{2n}$,

$$\begin{aligned} \left\| \widetilde{\xi}_o(t) \right\|_2 &\leq \sqrt{p_M V_o(t)} \\ &\leq \sqrt{\frac{p_M}{p_m}} \left\| \widetilde{\xi}_o(0) \right\|_2 + \frac{2}{\lambda_s} \sqrt{\frac{p_M}{p_m}}\, \bar{w}_o = \bar{\xi}_o. \end{aligned}$$

□

D. Relative Tracking Control and Closed-Loop Analysis

Using the relative-localization estimator (51), the pursuer can stably estimate the target's relative position even when the target employs an escape strategy. Based on this estimate, we design the following controller to achieve effective target tracking.

$$u_O = \widehat{u}_T + \widehat{\delta}_v - \dot{r}_d + K_r(\widehat{r} - r_d), \qquad K_r = K_r^\top > 0. \tag{63}$$

Define the tracking error $e_r = r - r_d$.

To guarantee the stability of the proposed tracking controller (63), the following assumption should be satisfied

Assumption 3. *The desired relative trajectory $r_d(t) \in \mathbb{R}^n$ is continuously differentiable and satisfies*

$$\|r_d(t)\|_2 \leq \bar{r}_d, \qquad \|\dot{r}_d(t)\|_2 \leq \bar{r}_{d,1},$$

where $\bar{r}_d > 0$ and $\bar{r}_{d,1} > 0$ are finite constants.

The stability of the proposed tracking controller (63) can be described in the following theorem.

Theorem 1. *Suppose that Assumptions 1, 3, and 2 and the conditions of Lemma 1 hold. Consider the identifier* (24)*, the filter* (46)*, the localization estimator* (50)–(51)*, and the controller* (63)*. Choose*

$$\mu > \frac{\left(\|K_r\|_2 + 1\right)^2 p_M}{\lambda_s \lambda_{\min}(K_r)}$$

and define

$$\gamma_\mu = \min\left\{\lambda_{\min}(K_r), \frac{\lambda_s}{2} - \frac{\left(\|K_r\|_2 + 1\right)^2 p_M}{2\mu\lambda_{\min}(K_r)}\right\} > 0. \quad (64)$$

Furthermore, let

$$\bar{e}_r = \left(\|e_r(0)\|_2^2 + \frac{2\mu}{p_m}\left\|\widetilde{\xi}_o(0)\right\|_2^2 + \frac{4\mu}{\gamma_\mu \lambda_s p_m}\bar{w}_o^2\right)^{1/2}.$$

Then

$$\|e_r(t)\|_2 \leq \bar{e}_r, \qquad \forall t \geq 0. \quad (65)$$

Moreover, the parameter, localization, and tracking errors are uniformly ultimately bounded, with

$$\begin{aligned}
\limsup_{t\to\infty}\left\|\widetilde{\theta}(t)\right\|_2 &\leq \frac{\bar{d}_\theta}{c_\theta}, \\
\limsup_{t\to\infty}\left\|\widetilde{\xi}_o(t)\right\|_2 &\leq \frac{2}{\lambda_s}\sqrt{\frac{p_M}{p_m}}\,\bar{w}_o, \\
\limsup_{t\to\infty}\|e_r(t)\|_2 &\leq \sqrt{\frac{4\mu}{\gamma_\mu\lambda_s p_m}}\,\bar{w}_o.
\end{aligned} \quad (66)$$

Proof. From (48) and (63),

$$\dot{e}_r = -K_r e_r + K_r\widetilde{r} + \widetilde{\delta}_v.$$

Let

$$V_r = \frac{1}{2}\|e_r\|_2^2, \qquad V_c = V_r + \mu V_o.$$

Since $\widetilde{\xi}_o = \mathrm{col}\{\widetilde{r}, \widetilde{\delta}_v\}$,

$$\left\|K_r\widetilde{r} + \widetilde{\delta}_v\right\|_2 \leq \left(\|K_r\|_2 + 1\right)\left\|\widetilde{\xi}_o\right\|_2.$$

Consequently,

$$\begin{aligned}
\dot{V}_r \leq & -\lambda_{\min}(K_r)\|e_r\|_2^2 + \left(\|K_r\|_2 + 1\right)\|e_r\|_2\left\|\widetilde{\xi}_o\right\|_2 \\
\leq & -\lambda_{\min}(K_r)V_r + \frac{\left(\|K_r\|_2 + 1\right)^2 p_M}{2\lambda_{\min}(K_r)}V_o.
\end{aligned} \quad (67)$$

From the proof of Lemma 4,

$$\dot{V}_o \leq -\frac{\lambda_s}{2}V_o + \frac{2}{\lambda_s p_m}\bar{w}_o^2. \quad (68)$$

Combining (67) and (68), and using (64), gives

$$\dot{V}_c \leq -\gamma_\mu V_c + \frac{2\mu}{\lambda_s p_m}\bar{w}_o^2.$$

The comparison Lemma yields

$$V_c(t) \leq e^{-\gamma_\mu t}V_c(0) + \frac{2\mu}{\gamma_\mu\lambda_s p_m}\left(1 - e^{-\gamma_\mu t}\right)\bar{w}_o^2. \quad (69)$$

Since $\|e_r\|_2^2 = 2V_r \leq 2V_c$,

$$\|e_r(t)\|_2^2 \leq \|e_r(0)\|_2^2 + \frac{2\mu}{p_m}\left\|\widetilde{\xi}_o(0)\right\|_2^2 + \frac{4\mu}{\gamma_\mu\lambda_s p_m}\bar{w}_o^2.$$

This proves (65). The three ultimate bounds in (66) follow from (25), (62), and (69), respectively. □

IV. Conclusion

The conclusion goes here.

Appendix A
Proof of the First Zonklar Equation

Appendix one text goes here.

Appendix B

Appendix two text goes here.

References

[1] H. Kopka and P. W. Daly, *A Guide to LaTeX*, 3rd ed. Harlow, England: Addison-Wesley, 1999.

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Michael Shell Biography text here.

John Doe Biography text here.

Jane Doe Biography text here.